Chapter 7. Stokes' Theorem

Integrating forms over manifolds, orientations, the generalized Stokes' theorem and its applications to vector analysis

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Overview

Chapter 6 showed how k-forms generalize scalar and vector fields in $\mathbb{R}^3$, and how d generalizes grad, curl and div. We now define the integral of a k-form over a k-manifold, generalizing line and surface integrals in $\mathbb{R}^3$; these appear in the classical Stokes' theorem and divergence theorem, and integrals of forms appear in the generalized Stokes' theorem (§37). Recall the convention: all manifolds, forms, vector fields and scalar fields are of class C^∞.

§33. Integrating Forms over Parametrized-Manifolds

Definition 33.1 (integral of a k-form over an open set of $\mathbb{R}^k$)

Let A be open in $\mathbb{R}^k$ and η a k-form defined in A; η can be written uniquely as $\eta = f\,dx_1 \wedge \cdots \wedge dx_k$. The **integral of η over A** is

$$\int_A \eta = \int_A f,$$

provided the latter integral exists.

This seems coordinate-dependent, since dx_i depends on the standard basis of $\mathbb{R}^k$. But if $\mathbf{a}_1, \ldots, \mathbf{a}_k$ is *any* right-handed orthonormal basis of $\mathbb{R}^k$, one checks $\int_A \eta = \int_{\mathbf{x}\in A} \eta(\mathbf{x})((\mathbf{x};\mathbf{a}_1), \ldots, (\mathbf{x};\mathbf{a}_k))$; so the integral depends on the orientation of $\mathbb{R}^k$ but not on the choice of basis.

Definition 33.2 (integral over a parametrized-manifold)

Let A be open in $\mathbb{R}^k$, $\alpha : A \to \mathbb{R}^n$ of class C^∞, and $Y = \alpha(A)$, so that Y_α is a parametrized-manifold. If ω is a k-form defined in an open set of $\mathbb{R}^n$ containing Y, the **integral of ω over Y_α** is

$$\int_{Y_\alpha} \omega = \int_A \alpha^*\omega,$$

provided the latter integral exists. It is linear in ω.

The integral is invariant under reparametrization *up to sign*:

Theorem 33.1

Let $g : A \to B$ be a diffeomorphism of open sets in $\mathbb{R}^k$. Assume $\det Dg$ does not change sign on A. Let $\beta : B \to \mathbb{R}^n$ be a map of class C^∞; let $Y = \beta(B)$. Let $\alpha = \beta \circ g$; then $\alpha : A \to \mathbb{R}^n$ and $Y = \alpha(A)$. If ω is a k-form defined in an open set of $\mathbb{R}^n$ containing Y, then ω is integrable over Y_β if and only if it is integrable over Y_α; in this case,

$$\int_{Y_\alpha} \omega = \pm \int_{Y_\beta} \omega,$$

where the sign agrees with the sign of $\det Dg$.

Proof. Let $\mathbf{x}$, $\mathbf{y}$ denote the general points of A, B. We must show $\int_A \alpha^*\omega = \epsilon \int_B \beta^*\omega$, with $\epsilon = \pm 1$ the sign of $\det Dg$. Put $\eta = \beta^*\omega$; since $\alpha^*\omega = g^*(\beta^*\omega)$, we need $\int_A g^*\eta = \epsilon \int_B \eta$. Write $\eta = f\,dy_1 \wedge \cdots \wedge dy_k$. By Theorem 32.2 (with $k = n$),

$$g^*\eta = (f \circ g)g^*(dy_1 \wedge \cdots \wedge dy_k) = (f \circ g)\det(Dg)\,dx_1 \wedge \cdots \wedge dx_k,$$

so we need $\int_A (f \circ g) \det Dg = \epsilon \int_B f$. This is the change of variables theorem, since $\det Dg = \epsilon |\det Dg|$. ■

If A is connected, the hypothesis that $\det Dg$ does not change sign is automatic: $\{\det Dg > 0\}$ and $\{\det Dg < 0\}$ are disjoint open sets with union A.

Theorem 33.2

Let A be open in $\mathbb{R}^k$; let $\alpha : A \to \mathbb{R}^n$ be of class C^∞; let $Y = \alpha(A)$. Let $\mathbf{x}$ denote the general point of A; and let $\mathbf{z}$ denote the general point of $\mathbb{R}^n$. If $\omega = f\, dz_I$ is a k-form defined in an open set of $\mathbb{R}^n$ containing Y, then

$$\int_{Y_\alpha} \omega = \int_A (f \circ \alpha) \det(\partial \alpha_I / \partial \mathbf{x}).$$

Proof. By Theorem 32.2, $\alpha^* \omega = (f \circ \alpha) \det(\partial \alpha_I / \partial \mathbf{x})\, dx_1 \wedge \cdots \wedge dx_k$. ■

The notion of a k-form, and even more of its integral over a parametrized-manifold, is abstract; §36 gives a geometric interpretation.

The "dx" notation of calculus

If $\eta = f\, dx$ is a 1-form on $A = (a, b)$, then $\int_A \eta = \int_A f$, i.e. $\int_A f\, dx = \int_a^b f$: the left side is the integral of a *form*, the right of a *function*, and they are equal by definition. Line integrals also make sense: for a 1-form $P\, dx + Q\, dy + R\, dz$ on an open set of $\mathbb{R}^3$ and a parametrized-curve $\gamma : (a, b) \to \mathbb{R}^3$ with image C, Theorem 33.2 gives

$$\int_{C_\gamma} P\, dx + Q\, dy + R\, dz = \int_{(a,b)} \Big[P(\gamma(t)) \frac{d\gamma_1}{dt} + Q(\gamma(t)) \frac{d\gamma_2}{dt} + R(\gamma(t)) \frac{d\gamma_3}{dt} \Big],$$

the calculus formula. The "$dx\, dy$" notation for double integrals is harder: $\iint_A f(x, y)\, dx\, dy$ resembles both the iterated integral, $\int_c^d [\int_a^b f(x, y)\, dx]\, dy = \int_A f$, and the integral of a 2-form, $\int_A f\, dx \wedge dy = \int_A f$. Reversing x and y, the iterated integral still gives $\int_A f$ but $\int_A f\, dy \wedge dx = -\int_A f$. To avoid confusion we do not use "$dx\, dy$". The "dV" notation of Chapter 5 is unambiguous: if A is open in $\mathbb{R}^k$, parametrized by the identity, then $\int_{A_\alpha} f\, dV = \int_A (f \circ \alpha) V(D\alpha) = \int_A f$; the d there has nothing to do with the operator d.

§34. Orientable Manifolds

For ω supported in a single patch $\alpha : U \to V$ on M we want $\int_M \omega = \int_{\mathrm{Int}\, U} \alpha^* \omega$. Since this is invariant under reparametrization only up to sign, we need an extra condition on M: *orientability*.

Definition 34.1 (orientation-preserving diffeomorphism)

A diffeomorphism $g : A \to B$ of open sets in $\mathbb{R}^k$ is **orientation-preserving** if $\det Dg > 0$ on A, and **orientation-reversing** if $\det Dg < 0$ on A.

This generalizes §20: g is orientation-preserving iff for each $\mathbf{x}$ the linear transformation of $\mathbb{R}^k$ with matrix $Dg(\mathbf{x})$, i.e. $g_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_{g(\mathbf{x})}(\mathbb{R}^k)$, is orientation-preserving.

Definition 34.2 (orientable manifold, orientation)

Let M be a k-manifold in $\mathbb{R}^n$. Coordinate patches $\alpha_i : U_i \to V_i$ $(i = 0, 1)$ on M **overlap positively** if the transition function $\alpha_1^{-1} \circ \alpha_0$ is orientation-preserving (when $V_0 \cap V_1 \neq \emptyset$). If M can be covered by coordinate patches each pair of which overlap positively (if they overlap at all), M is **orientable**; otherwise **non-orientable**. Given such a covering, adjoin all other patches overlapping these positively; the expanded collection still overlaps positively and is an **orientation** on M. A manifold with an orientation is an **oriented manifold**.

This makes no sense for 0-manifolds; see §37. For a k-dimensional vector space V we now have two notions of orientation; they correspond: given an orientation of V in the sense of §20, for each frame $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ in it the linear isomorphism $\alpha : \mathbb{R}^k \to V$ with $\alpha(\mathbf{e}_i) = \mathbf{v}_i$ is a coordinate patch on V; any two such overlap positively, and they specify an orientation in the present sense.

34.1 Oriented manifolds in $\mathbb{R}^n$ of dimensions 1, $n-1$ and n

Definition 34.3 (unit tangent field)

Let M be an oriented 1-manifold in $\mathbb{R}^n$. For $\mathbf{p} \in M$ choose a patch $\alpha : U \to V$ about $\mathbf{p}$ in the orientation, with $\alpha(t_0) = \mathbf{p}$, and define

$$T(\mathbf{p}) = (\mathbf{p}; D\alpha(t_0)/\|D\alpha(t_0)\|).$$

T is the **unit tangent field corresponding to the orientation** of M.

$(\mathbf{p}; D\alpha(t_0))$ is the velocity of the curve α at t_0, and $T(\mathbf{p})$ is this vector normalized. T is well-defined: if β is another patch in the orientation with $\mathbf{p} = \beta(t_1)$ and $g = \beta^{-1} \circ \alpha$, then $D\alpha(t_0) = D\beta(t_1) \cdot Dg(t_0)$ with $Dg(t_0) > 0$ (orientation-preserving), so $D\alpha(t_0)/\|D\alpha(t_0)\| = D\beta(t_1)/\|D\beta(t_1)\|$. T is C^∞ since $t_0 = \alpha^{-1}(\mathbf{p})$ is a C^∞ function of $\mathbf{p}$.

Example 34.1

An oriented 1-manifold M with unit tangent field T is pictured by drawing arrows along M in the direction of T; this is a *directed curve* in calculus.

If M has boundary points $\mathbf{p}, \mathbf{q}$ (e.g. an arc), a patch $\alpha : U \to V$ about the boundary point $\mathbf{p}$ has U open in $\mathbb{H}^1$, so $T(\mathbf{p})$ must point *into* M from $\mathbf{p}$; likewise at $\mathbf{q}$. Then no unit tangent field points into M at both ends, and M would seem non-orientable—an anomaly. It disappears if we also allow patches with domains open in the left half-line $\mathbb{L}^1 = \{x : x \leq 0\}$.

Convention

In the case of a 1-manifold M, we allow the domains of the coordinate patches on M to be open sets in $\mathbb{R}^1$ or in $\mathbb{H}^1$ or in $\mathbb{L}^1$.

With this, every 1-manifold is orientable (we shall not prove this).

Definition 34.4 (unit normal field)

Let M be an $(n-1)$-manifold in $\mathbb{R}^n$ and $\mathbf{p} \in M$. Let $(\mathbf{p}; \mathbf{n})$ be a unit vector in $\mathcal{T}_\mathbf{p}(\mathbb{R}^n)$ orthogonal to the $(n-1)$-dimensional subspace $\mathcal{T}_\mathbf{p}(M)$; $\mathbf{n}$ is determined up to sign. Given an orientation of M, choose a patch $\alpha : U \to V$ in it with $\alpha(\mathbf{x}) = \mathbf{p}$; the columns $\partial\alpha/\partial x_i$ give a basis of $\mathcal{T}_\mathbf{p}(M)$. Fix the sign of $\mathbf{n}$ by requiring the frame $(\mathbf{n}, \partial\alpha/\partial x_1, \dots, \partial\alpha/\partial x_{n-1})$ to be right-handed, i.e. $\det[\mathbf{n}\ D\alpha(\mathbf{x})] > 0$. The field $N(\mathbf{p}) = (\mathbf{p}; \mathbf{n}(\mathbf{p}))$ is the **unit normal field**

to M corresponding to the orientation of M. (It is well-defined and C^∞; see §38.)

Example 34.2 (Möbius band)

The Möbius band in $\mathbb{R}^3$ has no continuous unit normal vector field; it is not orientable.

Example 34.3 (Klein bottle)

The Klein bottle K can be pictured in $\mathbb{R}^3$ as a self-intersecting surface swept out by a moving circle $C_0, C_1, C_2, \ldots$. It can be represented as a 2-manifold *without* self-intersections in $\mathbb{R}^4$: keep the circle in $\mathbb{R}^3 \times 0$, but as it approaches the spot where it would cross part of the surface already generated, let it move "up" into $\mathbb{R}^3 \times \mathbb{H}^1_+$ until past that spot, then come back down. K contains a copy of the Möbius band M; if K were orientable, so would be M (take all patches on M that overlap positively the patches of the orientation of K). So K is not orientable.

Definition 34.5 (natural orientation)

Let M be an n-manifold in $\mathbb{R}^n$. For a patch $\alpha : U \to V$ on M, $D\alpha$ is n by n. The **natural orientation** of M consists of all coordinate patches α on M with $\det D\alpha > 0$; any two overlap positively.

M can be covered by such patches: given $\mathbf{p}$ and a patch $\alpha : U \to V$ about $\mathbf{p}$ with U open in $\mathbb{R}^n$ or $\mathbb{H}^n$, shrink U to an open ϵ-ball or its intersection with $\mathbb{H}^n$; then U is connected and $\det D\alpha$ has constant sign on U. If negative, replace α by $\alpha \circ r$ with $r(x_1, \ldots, x_n) = (-x_1, x_2, \ldots, x_n)$.

34.2 Reversing the orientation of a manifold

Definition 34.6 (reverse orientation)

Let $r : \mathbb{R}^k \to \mathbb{R}^k$ be the reflection $r(x_1, \ldots, x_k) = (-x_1, x_2, \ldots, x_k)$; it is its own inverse, carries $\mathbb{H}^k$ to itself if $k > 1$, and $\mathbb{H}^1$ to $\mathbb{L}^1$ if $k = 1$. If M is an oriented k-manifold and $\alpha_i : U_i \to V_i$ are the patches of its orientation, let $\beta_i = \alpha_i \circ r : r(U_i) \to V_i$. Then β_i overlaps α_i negatively, while the β_i overlap each other positively; they form the **reverse** (or **opposite**) orientation.

So every orientable k-manifold has at least two orientations; a connected one has exactly two. (A 1-manifold with four components, for example, has four orientations.) If M is an oriented 1-manifold with tangent field T, reversing the orientation replaces T by $-T$, since $(\alpha \circ r)(t) = \alpha(-t)$. If M is an oriented $(n-1)$-manifold with normal field N, reversing replaces N by $-N$: $\partial(\alpha \circ r)/\partial x_1 = -\partial\alpha/\partial x_1$, $\partial(\alpha \circ r)/\partial x_i = \partial\alpha/\partial x_i$ for $i > 1$, and $(\mathbf{n}, \partial\alpha/\partial x_1, \ldots)$ is right-handed iff $(-\mathbf{n}, -\partial\alpha/\partial x_1, \partial\alpha/\partial x_2, \ldots)$ is.

34.3 The induced orientation of ∂M

Theorem 34.1

Let $k > 1$. If M is an orientable k-manifold with non-empty boundary, then ∂M is orientable.

Proof. Let $\mathbf{p} \in \partial M$ and $\alpha : U \to V$ a patch about $\mathbf{p}$. The restricted patch on ∂M (§24) is $\alpha_0 = \alpha \circ b$, where $b : \mathbb{R}^{k-1} \to \mathbb{R}^k$, $b(x_1, \ldots, x_{k-1}) = (x_1, \ldots, x_{k-1}, 0)$. We show: if α, β are patches about $\mathbf{p}$

overlapping positively, then so do α_0, β_0. Let $g = \beta^{-1} \circ \alpha : W_0 \to W_1$, with W_0, W_1 open in $\mathbb{H}^k$; then $\det Dg > 0$. If $\mathbf{x} \in \partial\mathbb{H}^k$, the last row of $Dg(\mathbf{x})$ is $Dg_k = [0 \ \cdots \ 0 \ \partial g_k/\partial x_k]$ with $\partial g_k/\partial x_k \geq 0$: incrementing one of $x_1, \ldots, x_{k-1}$ from a point of $\partial\mathbb{H}^k$ does not change g_k (which stays 0, as g maps $\partial\mathbb{H}^k$ into $\partial\mathbb{H}^k$), while incrementing x_k positively increases g_k (as g maps into $\mathbb{H}^k$). Since $\det Dg \neq 0$, $\partial g_k/\partial x_k > 0$, and since $\det Dg > 0$,

$$\det \frac{\partial(g_1, \ldots, g_{k-1})}{\partial(x_1, \ldots, x_{k-1})} > 0.$$

This matrix is the derivative of the transition function $\beta_0^{-1} \circ \alpha_0$. ■

So an orientation of M gives one of ∂M by restricting the patches of the orientation; but this is not always the one we prefer:

Definition 34.7 (induced orientation of ∂M)

Let M be an orientable k-manifold with non-empty boundary and fix an orientation of M. The **induced orientation** of ∂M is: if k is even, the orientation obtained by restricting the patches of the orientation of M; if k is odd, the opposite of that orientation.

Example 34.4

S^2 and the torus T are orientable 2-manifolds, being boundaries of orientable 3-manifolds in $\mathbb{R}^3$. If M is a 3-manifold in $\mathbb{R}^3$ oriented naturally, the induced orientation of ∂M corresponds to the unit normal field pointing *outwards* from M (proved in §38). Informally: let α be a patch about $\mathbf{p} \in \partial M$ in the natural orientation; $(\alpha \circ b)(\mathbf{x}) = \alpha(x_1, x_2, 0)$ is the restricted patch. Since $\dim M = 3$ is odd, the induced orientation of ∂M is opposite to that of the restrictions, so the normal $N = (\mathbf{p}; \mathbf{n})$ for the induced orientation satisfies: $(-\mathbf{n}, \partial\alpha/\partial x_1, \partial\alpha/\partial x_2)$ is right-handed. Since $\det D\alpha > 0$, $(\partial\alpha/\partial x_3, \partial\alpha/\partial x_1, \partial\alpha/\partial x_2)$ is right-handed. So $-\mathbf{n}$ and $\partial\alpha/\partial x_3$ lie on the same side of the tangent plane. Since $\partial\alpha/\partial x_3$ points *into* M, $\mathbf{n}$ points outwards.

Example 34.5

Let M be an oriented 2-manifold in $\mathbb{R}^3$ with non-empty boundary, N the unit normal field for the orientation of M, and T the unit tangent field for the induced orientation of ∂M. For $\mathbf{p} \in \partial M$ let $W(\mathbf{p})$ be the unit vector perpendicular to $N(\mathbf{p})$ and $T(\mathbf{p})$ with (N, T, W) right-handed. Then $W(\mathbf{p})$ is tangent to M and points into M. (This makes precise the calculus description: walking around ∂M in the direction of T with head pointing along N, M is on your left.) Indeed, let $\alpha : U \to V$ be a patch about $\mathbf{p}$ in the orientation of M; since $\dim M = 2$ is even, $\alpha \circ b$ belongs to the induced orientation of ∂M, so $\partial\alpha/\partial x_1$ points in the direction of T. $\partial\alpha/\partial x_2$ is the velocity of a curve moving into M, so it points into M; choose λ with $\mathbf{w} = \partial\alpha/\partial x_2 + \lambda\,\partial\alpha/\partial x_1$ orthogonal to $\partial\alpha/\partial x_1$; then $\mathbf{w}$ points into M, and $W(\mathbf{p}) = (\mathbf{p}; \mathbf{w}/\|\mathbf{w}\|)$. By definition $(\mathbf{n}, \partial\alpha/\partial x_1, \partial\alpha/\partial x_2)$ is right-handed, and $\det[\mathbf{n}\ \partial\alpha/\partial x_1\ \partial\alpha/\partial x_2] = \det[\mathbf{n}\ \partial\alpha/\partial x_1\ \mathbf{w}]$; so (N, T, W) is right-handed.

§35. Integrating Forms over Oriented Manifolds

The procedure parallels §25, so we abbreviate the details.

Definition 35.1 (integral, support in one patch)

Let M be a compact oriented k-manifold in $\mathbb{R}^n$ and ω a k-form defined in an open set of $\mathbb{R}^n$ containing M. Let $C = M \cap (\operatorname{Support}\omega)$, a compact set. Suppose there is a coordinate patch $\alpha : U \to V$ on M belonging to the orientation of M with $C \subset V$; we may assume U bounded. Define

$$\int_M \omega = \int_{\operatorname{Int} U} \alpha^*\omega,$$

where $\operatorname{Int} U = U$ if U is open in $\mathbb{R}^k$ and $\operatorname{Int} U = U \cap \mathbb{H}^k_+$ if U is open in $\mathbb{H}^k$ but not in $\mathbb{R}^k$.

Existence: α extends to a C^∞ map on an open set U' of $\mathbb{R}^k$, and $\alpha^*\omega$ extends to a C^∞ form $h\,dx_1 \wedge \cdots \wedge dx_k$ on U'; then $\int_{\operatorname{Int} U} \alpha^*\omega = \int_{\operatorname{Int} U} h$. h is continuous on U and vanishes outside the compact set $\alpha^{-1}(C)$, so it is bounded; it vanishes near each point of $\operatorname{Bd} U$ (if U is open in $\mathbb{R}^k$), or near each point of $\operatorname{Bd} U$ not in $\partial\mathbb{H}^k$, a set of measure zero. So h is integrable over U and over $\operatorname{Int} U$. *Well-definedness*: as in Lemma 25.1, using now that transition functions between patches of the orientation are orientation-preserving, so the sign in Theorem 33.1 is $+$. *Linearity*: if ω and η have supports whose intersections with M lie in a single patch of the orientation, $\int_M a\omega + b\eta = a\int_M \omega + b\int_M \eta$, since α^* and $\int_{\operatorname{Int} U}$ are linear. *Orientation*: if $-M$ denotes M with the opposite orientation, $\int_{-M} \omega = -\int_M \omega$, by Theorem 33.1.

Definition 35.2 (integral of a form over a compact oriented manifold)

Let M be a compact oriented k-manifold in $\mathbb{R}^n$ and ω a k-form defined in an open set of $\mathbb{R}^n$ containing M. Cover M by coordinate patches belonging to its orientation, and choose a partition of unity $\phi_1, \ldots, \phi_l$ on M dominated by this collection (Lemma 25.2). Define

$$\int_M \omega = \sum_{i=1}^{l} \Big[\int_M \phi_i\omega\Big].$$

This agrees with Definition 35.1 when $M \cap \operatorname{Support}\omega$ lies in one patch, by linearity and $\omega(\mathbf{x}) = \sum_i \phi_i(\mathbf{x})\omega(\mathbf{x})$ for $\mathbf{x} \in M$; it is independent of the partition of unity by the argument used for $\int_M f\,dV$.

Theorem 35.1

Let M be a compact oriented k-manifold in $\mathbb{R}^n$. Let ω, η be k-forms defined in an open set of $\mathbb{R}^n$ containing M. Then

$$\int_M (a\omega + b\eta) = a\int_M \omega + b\int_M \eta.$$

If $-M$ denotes M with the opposite orientation, then $\int_{-M} \omega = -\int_M \omega$.

Proof (supplied). Apply the single-patch facts above to each $\phi_i\omega$, $\phi_i\eta$ and sum; for $-M$, the patches $\alpha_i \circ r$ of the reverse orientation and the same partition of unity may be used. ■

As with $\int_M f\,dV$, in practice one evaluates $\int_M \omega$ by breaking M into pieces:

Theorem 35.2

Let M be a compact oriented k-manifold in $\mathbb{R}^n$. Let ω be a k-form defined in an open set of $\mathbb{R}^n$ containing M. Suppose that $\alpha_i : A_i \to M_i$, for $i = 1, \ldots, N$, is a coordinate patch on M belonging to the orientation of M, such that A_i is open in $\mathbb{R}^k$ and M is the disjoint union of

the open sets $M_1, \dots, M_N$ of M and a set K of measure zero in M. Then

$$\int_M \omega = \sum_{i=1}^{N} \Big[\int_{A_i} \alpha_i^* \omega \Big].$$

Proof (supplied). Copy the proof of Theorem 25.4: by linearity assume $M \cap \operatorname{Support} \omega$ lies in one patch $\alpha : U \to V$ of the orientation, and write $\alpha^*\omega = F\, dx_1 \wedge \cdots \wedge dx_k$, $\alpha_i^*\omega = F_i\, dx_1 \wedge \cdots \wedge dx_k$. With $W_i = \alpha^{-1}(M_i \cap V)$ and $L = \alpha^{-1}(K \cap V)$, $\int_M \omega = \int_{\operatorname{Int} U} F = \sum_i \int_{W_i} F$ as in Step 1 there. The transition map $\alpha_i^{-1} \circ \alpha : W_i \to B_i = \alpha_i^{-1}(M_i \cap V)$ is an orientation-preserving diffeomorphism, so by Theorem 33.1 (sign +) $\int_{W_i} F = \int_{B_i} F_i$, and $\int_{B_i} F_i = \int_{A_i} F_i$ by additivity as in Step 2 there. (Alternatively combine Theorems 25.4 and 36.2.) ■

§36. A Geometric Interpretation of Forms and Integrals

The integral of a k-form over an oriented k-manifold is abstract; we relate it to the integral of a scalar function with respect to volume. First, alternating tensors versus volume:

Theorem 36.1

Let W be a k-dimensional linear subspace of $\mathbb{R}^n$; let $(\mathbf{a}_1, \dots, \mathbf{a}_k)$ be an orthonormal k-frame in W, and let f be an alternating k-tensor on W. If $(\mathbf{x}_1, \dots, \mathbf{x}_k)$ is an arbitrary k-tuple in W, then

$$f(\mathbf{x}_1, \dots, \mathbf{x}_k) = \epsilon\, V(\mathbf{x}_1, \dots, \mathbf{x}_k)\, f(\mathbf{a}_1, \dots, \mathbf{a}_k),$$

where $\epsilon = \pm 1$. If the $\mathbf{x}_i$ are independent, then $\epsilon = +1$ if the frames $(\mathbf{x}_1, \dots, \mathbf{x}_k)$ and $(\mathbf{a}_1, \dots, \mathbf{a}_k)$ belong to the same orientation of W and $\epsilon = -1$ otherwise.

(If the $\mathbf{x}_i$ are dependent, $V(\mathbf{x}_1, \dots, \mathbf{x}_k) = 0$ by Theorem 21.3 and ϵ does not matter.)

Proof.

Step 1. $W = \mathbb{R}^k$. Then f is a multiple of the determinant: $f(\mathbf{x}_1, \dots, \mathbf{x}_k) = c \det[\mathbf{x}_1 \cdots \mathbf{x}_k]$. If the $\mathbf{x}_i$ are dependent, f vanishes. Otherwise $f(\mathbf{x}_1, \dots, \mathbf{x}_k) = c\,\epsilon_1 V(\mathbf{x}_1, \dots, \mathbf{x}_k)$ with $\epsilon_1 = +1$ iff $(\mathbf{x}_1, \dots, \mathbf{x}_k)$ is right-handed, and $f(\mathbf{a}_1, \dots, \mathbf{a}_k) = c\,\epsilon_2 V(\mathbf{a}_1, \dots, \mathbf{a}_k) = c\,\epsilon_2$ similarly. So $f(\mathbf{x}_1, \dots, \mathbf{x}_k) = (\epsilon_1/\epsilon_2) V(\mathbf{x}_1, \dots, \mathbf{x}_k) f(\mathbf{a}_1, \dots, \mathbf{a}_k)$, and $\epsilon_1/\epsilon_2 = +1$ iff the two frames belong to the same orientation.

Step 2. In general choose an orthogonal transformation h of $\mathbb{R}^n$ carrying W onto $\mathbb{R}^k \times 0$, with inverse $k : \mathbb{R}^k \times 0 \to W$. Then k^*f is an alternating tensor on $\mathbb{R}^k \times 0$; $(h(\mathbf{x}_i))$ is a k-tuple and $(h(\mathbf{a}_i))$ an orthonormal frame in $\mathbb{R}^k \times 0$. By Step 1, $(k^*f)(h(\mathbf{x}_1), \dots, h(\mathbf{x}_k)) = \epsilon V(h(\mathbf{x}_1), \dots, h(\mathbf{x}_k))(k^*f)(h(\mathbf{a}_1), \dots, h(\mathbf{a}_k))$. Since V is unchanged by orthogonal transformations, this is the desired equation. If the $\mathbf{x}_i$ are independent, $\epsilon = +1$ iff $(h(\mathbf{x}_i))$, $(h(\mathbf{a}_i))$ belong to the same orientation of $\mathbb{R}^k \times 0$, iff $(\mathbf{x}_i)$, $(\mathbf{a}_i)$ belong to the same orientation of W. ■

So for two orthonormal frames $(\mathbf{a}_i)$, $(\mathbf{b}_i)$ in W, $f(\mathbf{a}_1, \dots, \mathbf{a}_k) = \pm f(\mathbf{b}_1, \dots, \mathbf{b}_k)$, the sign depending on whether they lie in the same orientation of W.

Definition 36.1 (natural orientation of $\mathcal{T}_{\mathbf{p}}(M)$)

Let M be an oriented k-manifold in $\mathbb{R}^n$ and $\mathbf{p} \in M$. Choose a patch $\alpha : U \to V$ in the orientation of M about $\mathbf{p}$, with $\alpha(\mathbf{x}) = \mathbf{p}$. The collection of k-frames in $\mathcal{T}_{\mathbf{p}}(M)$ of the form $(\alpha_*(\mathbf{x}; \mathbf{a}_1), \dots, \alpha_*(\mathbf{x}; \mathbf{a}_k))$, with $(\mathbf{a}_1, \dots, \mathbf{a}_k)$ right-handed in $\mathbb{R}^k$, is the **natural orientation** of $\mathcal{T}_{\mathbf{p}}(M)$ induced by the orientation of M; it is independent of α.

Theorem 36.2

Let M be a compact oriented k-manifold in $\mathbb{R}^n$; let ω be a k-form defined in an open set of $\mathbb{R}^n$ containing M. Let λ be the scalar function on M defined by the equation

$$\lambda(\mathbf{p}) = \omega(\mathbf{p})((\mathbf{p};\mathbf{a}_1),\dots,(\mathbf{p};\mathbf{a}_k)),$$

where $((\mathbf{p};\mathbf{a}_1),\dots,(\mathbf{p};\mathbf{a}_k))$ is any orthonormal frame in the linear space $\mathcal{T}_{\mathbf{p}}(M)$ belonging to its natural orientation. Then λ is continuous, and

$$\int_M \omega = \int_M \lambda\, dV.$$

Proof. By linearity assume $M \cap \operatorname{Support}\omega$ lies in one patch $\alpha : U \to V$ of the orientation. Write $\alpha^*\omega = h\, dx_1 \wedge \cdots \wedge dx_k$ and let $\alpha(\mathbf{x}) = \mathbf{p}$. Then

$$h(\mathbf{x}) = (\alpha^*\omega)(\mathbf{x})((\mathbf{x};\mathbf{e}_1),\dots,(\mathbf{x};\mathbf{e}_k)) = \omega(\mathbf{p})((\mathbf{p};\partial\alpha/\partial x_1),\dots,(\mathbf{p};\partial\alpha/\partial x_k)) = \pm V(D\alpha(\mathbf{x}))\lambda(\mathbf{p})$$

by Theorem 36.1; the sign is + because $((\mathbf{p};\partial\alpha/\partial x_1),\dots,(\mathbf{p};\partial\alpha/\partial x_k))$ belongs to the natural orientation of $\mathcal{T}_{\mathbf{p}}(M)$ by definition. Since $V(D\alpha) \neq 0$ ($D\alpha$ has rank k) and $\mathbf{x} = \alpha^{-1}(\mathbf{p})$ is continuous in $\mathbf{p}$, $\lambda(\mathbf{p}) = h(\mathbf{x})/V(D\alpha(\mathbf{x}))$ is continuous. Hence

$$\int_M \lambda\, dV = \int_{\operatorname{Int} U} (\lambda\circ\alpha)V(D\alpha) = \int_{\operatorname{Int} U} h = \int_{\operatorname{Int} U} \alpha^*\omega = \int_M \omega.$$

■

So for each k-form ω there is a scalar function λ (in fact C^∞) with $\int_M \omega = \int_M \lambda\, dV$. The converse is true but harder: there is a k-form ω_V, defined on an open set about M, whose value on any orthonormal basis of $\mathcal{T}_{\mathbf{p}}(M)$ in its natural orientation is 1. Then for any C^∞ function λ on M, $\int_M \lambda\, dV = \int_M \lambda\omega_V$, so the integral of λ over M is the integral of the form $\lambda\omega_V$. ω_V is a **volume form** for M, since $\int_M \omega_V = \int_M dV = \mathrm{v}(M)$. This applies only to orientable M: on a non-orientable M the integral of a scalar function is defined but that of a form is not.

Notation

Some mathematicians denote the volume form ω_V by dV (or dV). This makes the above equations tautologies, but it can confuse: V is *not* a form, and d does *not* denote the differential operator in this context.

§37. The Generalized Stokes' Theorem

This is the culmination: a theorem about integrals of forms containing the three basic theorems of vector integral calculus—Green's theorem, Stokes' theorem and the divergence theorem. Let $I^k = [0,1]^k$ be the unit k-cube; $\operatorname{Int} I^k = (0,1)^k$ and $\operatorname{Bd} I^k = I^k - \operatorname{Int} I^k$.

Lemma 37.1

Let $k > 1$. Let η be a $k-1$ form defined in an open set U of $\mathbb{R}^k$ containing the unit k-cube I^k. Assume η vanishes at all points of $\operatorname{Bd} I^k$ except possibly at points of the subset $(\operatorname{Int} I^{k-1})\times 0$. Then

$$\int_{\operatorname{Int} I^k} d\eta = (-1)^k \int_{\operatorname{Int} I^{k-1}} b^*\eta,$$

where $b : I^{k-1} \to I^k$ is the map $b(u_1,\dots,u_{k-1}) = (u_1,\dots,u_{k-1},0)$.

Proof. Let $\mathbf{x}$ and $\mathbf{u}$ be the general points of $\mathbb{R}^k$ and $\mathbb{R}^{k-1}$. For $1 \le j \le k$ let $I_j = (1, \dots, \hat{j}, \dots, k)$; the elementary $(k-1)$-forms in $\mathbb{R}^k$ are $dx_{I_j} = dx_1 \wedge \cdots \wedge \widehat{dx_j} \wedge \cdots \wedge dx_k$. By linearity of the integrals, d and b^*, it suffices to take $\eta = f\, dx_{I_j}$.

Step 1. We have

$$d\eta = df \wedge dx_{I_j} = \Big(\sum_{i=1}^{k} D_i f\, dx_i\Big) \wedge dx_{I_j} = (-1)^{j-1}(D_j f)\, dx_1 \wedge \cdots \wedge dx_k.$$

So, using the Fubini theorem with $\mathbf{v} = (x_1, \dots, \hat{x}_j, \dots, x_k)$,

$$\int_{\operatorname{Int} I^k} d\eta = (-1)^{j-1} \int_{I^k} D_j f = (-1)^{j-1} \int_{\mathbf{v} \in I^{k-1}} \int_{x_j \in I} D_j f(x_1, \dots, x_k).$$

By the fundamental theorem of calculus the inner integral is $f(x_1, \dots, 1, \dots, x_k) - f(x_1, \dots, 0, \dots, x_k)$, with 1 and 0 in the jth place. η, hence f, vanishes on $\operatorname{Bd} I^k$ except possibly on the open bottom face $(\operatorname{Int} I^{k-1}) \times 0$. If $j < k$ both terms vanish; if $j = k$ the result is $-f(x_1, \dots, x_{k-1}, 0)$. Hence

$$\int_{\operatorname{Int} I^k} d\eta = \begin{cases} 0 & j < k, \\ (-1)^k \int_{I^{k-1}} (f \circ b) & j = k. \end{cases}$$

Step 2. $Db = \begin{bmatrix} I_{k-1} \\ 0 \end{bmatrix}$, so by Theorem 32.2,

$$b^*(dx_{I_j}) = [\det Db(1, \dots, \hat{j}, \dots, k)]\, du_1 \wedge \cdots \wedge du_{k-1} = \begin{cases} 0 & j < k, \\ du_1 \wedge \cdots \wedge du_{k-1} & j = k. \end{cases}$$

Hence $\int_{\operatorname{Int} I^{k-1}} b^*\eta$ is 0 if $j < k$ and $\int_{\operatorname{Int} I^{k-1}} (f \circ b)$ if $j = k$. Compare with Step 1. ■

Theorem 37.2 (Stokes' theorem)

Let $k > 1$. Let M be a compact oriented k-manifold in $\mathbb{R}^n$; give ∂M the induced orientation if ∂M is not empty. Let ω be a $k-1$ form defined in an open set of $\mathbb{R}^n$ containing M. Then

$$\int_M d\omega = \int_{\partial M} \omega$$

if ∂M is not empty; and $\int_M d\omega = 0$ if ∂M is empty.

Proof.

Step 1. A special covering of M by coordinate patches. If $\mathbf{p} \in M - \partial M$, choose a patch $\alpha : U \to V$ in the orientation of M with U open in $\mathbb{R}^k$, $U \supset I^k$, and α carrying a point of $\operatorname{Int} I^k$ to $\mathbf{p}$ (precede an arbitrary patch by a translation and a stretching of $\mathbb{R}^k$). Let $W = \operatorname{Int} I^k$ and $Y = \alpha(W)$; then $\alpha : W \to Y$ is a coordinate patch in the orientation about $\mathbf{p}$, with W open in $\mathbb{R}^k$. If $\mathbf{p} \in \partial M$, choose $\alpha : U \to V$ in the orientation with U open in $\mathbb{H}^k$, $U \supset I^k$, carrying a point of $(\operatorname{Int} I^{k-1}) \times 0$ to $\mathbf{p}$. Let $W = (\operatorname{Int} I^k) \cup ((\operatorname{Int} I^{k-1}) \times 0)$ and $Y = \alpha(W)$; then $\alpha : W \to Y$ is a patch in the orientation about $\mathbf{p}$ with W open in $\mathbb{H}^k$ but not in $\mathbb{R}^k$. In each case α can be extended to a C^∞ map on an open set of $\mathbb{R}^k$ containing I^k.

Step 2. Since d and the integrals are linear, and $\omega = \sum_i \phi_i \omega$ for a partition of unity $\{\phi_i\}$ on M dominated by these patches, it suffices to prove the theorem when $C = M \cap \operatorname{Support} \omega$ is covered by a single patch $\alpha : W \to Y$ of Step 1. Then $M \cap \operatorname{Support} d\omega \subset C$ is covered too. Let $\eta = \alpha^*\omega$, extended (without change of notation) to a C^∞ form on an open set of $\mathbb{R}^k$ containing I^k. η vanishes at all points of $\operatorname{Bd} I^k$ except possibly points of $(\operatorname{Int} I^{k-1}) \times 0$, so Lemma 37.1 applies.

Step 3. Case $W = \operatorname{Int} I^k$; Y is disjoint from ∂M. Since $\alpha^* d\omega = d\alpha^*\omega = d\eta$,

$$\int_M d\omega = \int_{\operatorname{Int} I^k} \alpha^* d\omega = \int_{\operatorname{Int} I^k} d\eta = (-1)^k \int_{\operatorname{Int} I^{k-1}} b^*\eta.$$

Here η vanishes outside $\operatorname{Int} I^k$, in particular on $I^{k-1} \times 0$, so $b^*\eta = 0$ and $\int_M d\omega = 0$. If ∂M is empty, this is what we wanted. If not, $\int_{\partial M} \omega = 0$ too, since $\operatorname{Support} \omega$ is disjoint from ∂M.

Step 4. Case W open in $\mathbb{H}^k$, $Y \cap \partial M \neq \emptyset$. As before, $\int_M d\omega = \int_{\operatorname{Int} I^k} d\eta = (-1)^k \int_{\operatorname{Int} I^{k-1}} b^*\eta$. Now $\partial M \cap \operatorname{Support} \omega$ is covered by the patch $\beta = \alpha \circ b : \operatorname{Int} I^{k-1} \to Y \cap \partial M$ on ∂M, obtained by restricting α. β belongs to the induced orientation of ∂M if k is even, and to the opposite orientation if k is odd; so computing $\int_{\partial M} \omega$ with β we must reverse the sign when k is odd:

$$\int_{\partial M} \omega = (-1)^k \int_{\operatorname{Int} I^{k-1}} \beta^*\omega.$$

Since $\beta^*\omega = b^*(\alpha^*\omega) = b^*\eta$, the theorem follows. ■

What if $k = 1$? If ∂M is empty, one proves readily that $\int_M d\omega = 0$. If ∂M is non-empty we must say what an "orientation" of a 0-manifold is and how to "integrate" a 0-form over it. First a special case.

Definition 37.1 (arc)

Let M be a 1-manifold in $\mathbb{R}^n$. If there is a one-to-one C^∞ map $\alpha : [a, b] \to M$ carrying $[a, b]$ onto M with $D\alpha(t) \neq 0$ for all t, M is a (smooth) **arc** in $\mathbb{R}^n$. If M is oriented so that $\alpha|(a, b)$ belongs to the orientation, $\mathbf{p} = \alpha(a)$ is the **initial point** and $\mathbf{q} = \alpha(b)$ the **final point** of M.

Theorem 37.3

Let M be a 1-manifold in $\mathbb{R}^n$; let f be a 0-form defined in an open set about M. If M is an arc with initial point $\mathbf{p}$ and final point $\mathbf{q}$, then

$$\int_M df = f(\mathbf{q}) - f(\mathbf{p}).$$

Proof. $\alpha : (a, b) \to M - \mathbf{p} - \mathbf{q}$ is a patch covering all of M except a set of measure zero, so by Theorem 35.2, $\int_M df = \int_{(a,b)} \alpha^*(df)$. With t the general point of $\mathbb{R}$, $\alpha^*(df) = d(f \circ \alpha) = D(f \circ \alpha)\, dt$, so $\int_{(a,b)} \alpha^*(df) = \int_{(a,b)} D(f \circ \alpha) = f(\alpha(b)) - f(\alpha(a))$ by the fundamental theorem of calculus. ■

Definition 37.2 (orientation and integral for 0-manifolds)

A compact 0-manifold N in $\mathbb{R}^n$ is a finite set $\{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$. An **orientation** of N is a function $\epsilon : N \to \{-1, 1\}$. If f is a 0-form defined on an open set containing N, the **integral of f over the oriented manifold N** is

$$\int_N f = \sum_{i=1}^{m} \epsilon(\mathbf{x}_i) f(\mathbf{x}_i).$$

If M is an oriented 1-manifold with non-empty boundary, the **induced orientation** of ∂M is defined by $\epsilon(\mathbf{p}) = -1$ if there is a patch $\alpha : U \to V$ about $\mathbf{p}$ in the orientation of M with U open in $\mathbb{H}^1$, and $\epsilon(\mathbf{p}) = +1$ otherwise.

Theorem 37.4 (Stokes' theorem in dimension 1)

Let M be a compact oriented 1-manifold in $\mathbb{R}^n$; give ∂M the induced orientation if ∂M is not empty. Let f be a 0-form defined in an open set of $\mathbb{R}^n$ containing M. Then

$$\int_M df = \int_{\partial M} f$$

if ∂M is not empty; and $\int_M df = 0$ if ∂M is empty.

Proof (supplied). Cover M by patches $\alpha : W \to Y$ of its orientation, where W is one of the intervals $(0,1)$, $[0,1)$, $(-1,0]$, and use a partition of unity dominated by them; by linearity it suffices to treat f with $M \cap \operatorname{Support} f$ inside one such Y. Let $g = f \circ \alpha$, extended to a C^∞ function near $[-1,1]$; then $\int_M df = \int_{\operatorname{Int} W} \alpha^* df = \int_{\operatorname{Int} W} g'$. If $W = (0,1)$, g vanishes near 0 and 1, so $\int_0^1 g' = 0$, and $\operatorname{Support} f$ misses ∂M. If $W = [0,1)$, $\alpha(0) = \mathbf{p} \in \partial M$ has $\epsilon(\mathbf{p}) = -1$ and $g(1) = 0$: $\int_0^1 g' = -g(0) = -f(\mathbf{p}) = \int_{\partial M} f$. If $W = (-1,0]$, $\alpha(0) = \mathbf{p}$ has $\epsilon(\mathbf{p}) = +1$ (no patch of the orientation about $\mathbf{p}$ has domain open in $\mathbb{H}^1$, since such a patch would overlap α negatively), and $\int_{-1}^0 g' = g(0) = f(\mathbf{p}) = \int_{\partial M} f$. ■

§38. Applications to Vector Analysis

By §31, forms of order $k = 0, 1, n-1, n$ in $\mathbb{R}^n$ can be interpreted as scalar or vector fields. We show integrals of forms can be similarly interpreted, so that Stokes' theorem yields the classical theorems of vector integral calculus.

38.1 The gradient theorem for 1-manifolds in $\mathbb{R}^n$

Lemma 38.1

Let M be a compact oriented 1-manifold in $\mathbb{R}^n$; let T be the unit tangent vector to M corresponding to the orientation. Let $F(\mathbf{x}) = (\mathbf{x}; f(\mathbf{x})) = (\mathbf{x}; \sum f_i(\mathbf{x})\mathbf{e}_i)$ be a vector field defined in an open set of $\mathbb{R}^n$ containing M; it corresponds to the 1-form $\omega = \sum f_i\, dx_i$. Then

$$\int_M \omega = \int_M \langle F, T\rangle\, ds.$$

Here the classical ds (arc length) replaces dV. Replacing M by $-M$ changes the sign of $\int_M \omega$ and replaces T by $-T$, so the right side changes sign too.

Proof. *First proof.* By Theorem 36.2, $\int_M \omega = \int_M \lambda\, ds$, where $\lambda(\mathbf{p})$ is the value of $\omega(\mathbf{p})$ on an orthonormal basis of $\mathcal{T}_{\mathbf{p}}(M)$ in its natural orientation; here $T(\mathbf{p}) = (\mathbf{p}; \mathbf{t})$ is such a basis, and $\omega(\mathbf{p})(\mathbf{p}; \mathbf{t}) = \sum f_i(\mathbf{p}) t_i$. So $\lambda(\mathbf{p}) = \langle F(\mathbf{p}), T(\mathbf{p})\rangle$.

Second proof. By linearity assume $C = M \cap \operatorname{Support} \omega$ lies in a single patch $\alpha : U \to V$ of the orientation. With t the general point of $\mathbb{R}$, $\alpha^*\omega = \sum_i (f_i \circ \alpha)\, d\alpha_i = \sum_i (f_i \circ \alpha)(D\alpha_i)\, dt = \langle f \circ \alpha, D\alpha\rangle\, dt$, so $\int_M \omega = \int_{\operatorname{Int} U} \langle f \circ \alpha, D\alpha\rangle$. On the other hand

$$\int_M \langle F, T\rangle\, ds = \int_{\operatorname{Int} U} \langle F \circ \alpha, T \circ \alpha\rangle V(D\alpha) = \int_{\operatorname{Int} U} \langle f \circ \alpha, D\alpha / \|D\alpha\|\rangle V(D\alpha) = \int_{\operatorname{Int} U} \langle f \circ \alpha, D\alpha\rangle,$$

since $V(D\alpha) = [\det(D\alpha^{\mathrm{tr}} D\alpha)]^{1/2} = \|D\alpha\|$. ■

Theorem 38.2 (The gradient theorem)

Let M be a compact 1-manifold in $\mathbb{R}^n$; let T be a unit tangent vector field to M. Let f be a C^∞ function defined in an open set about M. If ∂M is empty, then $\int_M \langle \operatorname{grad} f, T\rangle\, ds = 0$. If ∂M consists of the points $\mathbf{x}_1, \dots, \mathbf{x}_m$, let $\epsilon_i = -1$ if T points into M at $\mathbf{x}_i$ and $\epsilon_i = +1$ otherwise. Then

$$\int_M \langle \operatorname{grad} f, T\rangle\, ds = \sum_{i=1}^{m} \epsilon_i f(\mathbf{x}_i).$$

Proof. The 1-form df corresponds to $\operatorname{grad} f$ (Theorem 31.1), so $\int_M df = \int_M \langle \operatorname{grad} f, T\rangle\, ds$ by the lemma. Apply Stokes' theorem in dimension 1. ■

38.2 The divergence theorem for $n-1$ manifolds in $\mathbb{R}^n$

Recall from §34 the unit normal field $N(\mathbf{p}) = (\mathbf{p}; \mathbf{n}(\mathbf{p}))$ of an oriented $(n-1)$-manifold M, defined by: $\mathbf{n}$ is a unit vector orthogonal to $\mathcal{T}_{\mathbf{p}}(M)$ and $(\mathbf{n}, \partial\alpha/\partial x_1, \dots, \partial\alpha/\partial x_{n-1})$ is right-handed for a patch α in the orientation with $\alpha(\mathbf{x}) = \mathbf{p}$. *$N$ is well-defined*: if β is another such patch, $g = \beta^{-1} \circ \alpha$ and $g(\mathbf{x}) = \mathbf{y}$, then $D\alpha(\mathbf{x}) = D\beta(\mathbf{y})Dg(\mathbf{x})$, so for any $\mathbf{v} \in \mathbb{R}^n$,

$$[\mathbf{v}\ D\alpha(\mathbf{x})] = [\mathbf{v}\ D\beta(\mathbf{y})] \cdot \begin{bmatrix} 1 & 0 \\ 0 & Dg(\mathbf{x}) \end{bmatrix}, \qquad \det[\mathbf{v}\ D\alpha(\mathbf{x})] = \det[\mathbf{v}\ D\beta(\mathbf{y})] \cdot \det Dg(\mathbf{x}).$$

Since $\det Dg > 0$, $[\mathbf{v}\ D\alpha(\mathbf{x})]$ has positive determinant iff $[\mathbf{v}\ D\beta(\mathbf{y})]$ does. To see N is C^∞ we find a formula for it.

Example 38.1

For $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$, calculus teaches that $\mathbf{c} = \mathbf{a} \times \mathbf{b}$ is perpendicular to both, $(\mathbf{c}, \mathbf{a}, \mathbf{b})$ is right-handed, and $\|\mathbf{c}\| = V(\mathbf{a}, \mathbf{b})$; here $c_1 = \det \begin{bmatrix} a_2 & b_2 \\ a_3 & b_3 \end{bmatrix}$, $c_2 = -\det \begin{bmatrix} a_1 & b_1 \\ a_3 & b_3 \end{bmatrix}$, $c_3 = \det \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}$. So for an oriented 2-manifold in $\mathbb{R}^3$ and a patch α in its orientation, $\mathbf{n} = \mathbf{c}/\|\mathbf{c}\|$ with $\mathbf{c} = \partial\alpha/\partial x_1 \times \partial\alpha/\partial x_2$ is the unit normal.

Lemma 38.3

Given independent vectors $\mathbf{x}_1, \dots, \mathbf{x}_{n-1}$ in $\mathbb{R}^n$, let X be the n by $n-1$ matrix $X = [\mathbf{x}_1\ \cdots\ \mathbf{x}_{n-1}]$, and let $\mathbf{c}$ be the vector $\mathbf{c} = \sum c_i \mathbf{e}_i$, where

$$c_i = (-1)^{i-1} \det X(1, \dots, \hat{\imath}, \dots, n).$$

The vector $\mathbf{c}$ has the following properties:
(1) $\mathbf{c}$ is non-zero and orthogonal to each $\mathbf{x}_i$.
(2) The frame $(\mathbf{c}, \mathbf{x}_1, \dots, \mathbf{x}_{n-1})$ is right-handed.
(3) $\|\mathbf{c}\| = V(X)$.

Proof. For $\mathbf{a} \in \mathbb{R}^n$, expanding by cofactors of the first column,

$$\det[\mathbf{a}\ \mathbf{x}_1\ \cdots\ \mathbf{x}_{n-1}] = \sum_{i=1}^{n} a_i (-1)^{i-1} \det X(1, \dots, \hat{\imath}, \dots, n) = \langle \mathbf{a}, \mathbf{c}\rangle.$$

(1) With $\mathbf{a} = \mathbf{x}_i$ the matrix has two equal columns, so $\langle \mathbf{x}_i, \mathbf{c}\rangle = 0$. Since the columns of X span an $(n-1)$-dimensional space, so do its rows; some $n-1$ rows are independent, say all but the ith; then $c_i \neq 0$.

(2) With $\mathbf{a} = \mathbf{c}$, $\det[\mathbf{c}\ \mathbf{x}_1 \cdots \mathbf{x}_{n-1}] = \|\mathbf{c}\|^2 > 0$.

(3) This follows from Theorem 21.4; alternatively,

$$[\mathbf{c}\ \mathbf{x}_1 \cdots \mathbf{x}_{n-1}]^{\mathrm{tr}} \cdot [\mathbf{c}\ \mathbf{x}_1 \cdots \mathbf{x}_{n-1}] = \begin{bmatrix} \|\mathbf{c}\|^2 & 0 \\ 0 & X^{\mathrm{tr}}X \end{bmatrix},$$

and taking determinants with (2), $\|\mathbf{c}\|^4 = \|\mathbf{c}\|^2 V(X)^2$. ■

Corollary 38.4

If M is an oriented $n-1$ manifold in $\mathbb{R}^n$, then the unit normal vector $N(\mathbf{p})$ corresponding to the orientation of M is a C^∞ function of $\mathbf{p}$.

Proof. For a patch $\alpha : U \to V$ in the orientation let $c_i(\mathbf{x}) = (-1)^{i-1} \det D\alpha(1, \ldots, \hat{\imath}, \ldots, n)(\mathbf{x})$ and $\mathbf{c}(\mathbf{x}) = \sum c_i(\mathbf{x})\mathbf{e}_i$. Then $N(\mathbf{p}) = (\mathbf{p}; \mathbf{c}(\mathbf{x})/\|\mathbf{c}(\mathbf{x})\|)$ with $\mathbf{x} = \alpha^{-1}(\mathbf{p})$, a C^∞ function of $\mathbf{p}$. ■

Lemma 38.5

Let M be a compact oriented $n-1$ manifold in $\mathbb{R}^n$; let N be the corresponding unit normal vector field. Let G be a vector field defined in an open set U of $\mathbb{R}^n$ containing M. If we denote the general point of $\mathbb{R}^n$ by $\mathbf{y}$, this vector field has the form $G(\mathbf{y}) = (\mathbf{y}; g(\mathbf{y})) = (\mathbf{y}; \sum g_i(\mathbf{y})\mathbf{e}_i)$; it corresponds to the $n-1$ form

$$\omega = \sum_{i=1}^{n} (-1)^{i-1} g_i\, dy_1 \wedge \cdots \wedge \widehat{dy_i} \wedge \cdots \wedge dy_n.$$

Then $\int_M \omega = \int_M \langle G, N \rangle\, dV$.

Replacing M by $-M$ changes the sign of $\int_M \omega$ and replaces N by $-N$.

Proof. *First proof.* By Theorem 36.2, $\int_M \omega = \int_M \lambda\, dV$, where $\lambda(\mathbf{p})$ is the value of $\omega(\mathbf{p})$ on an orthonormal basis $((\mathbf{p}; \mathbf{a}_1), \ldots, (\mathbf{p}; \mathbf{a}_{n-1}))$ of $\mathcal{T}_{\mathbf{p}}(M)$ in its natural orientation. Let $A = [\mathbf{a}_1 \cdots \mathbf{a}_{n-1}]$ and $\mathbf{c} = \sum c_i \mathbf{e}_i$ with $c_i = (-1)^{i-1} \det A(1, \ldots, \hat{\imath}, \ldots, n)$. By Lemma 38.3, $\mathbf{c}$ is orthogonal to each $\mathbf{a}_i$, $(\mathbf{c}, \mathbf{a}_1, \ldots, \mathbf{a}_{n-1})$ is right-handed, and $\|\mathbf{c}\| = V(A) = [\det(A^{\mathrm{tr}}A)]^{1/2} = [\det I_{n-1}]^{1/2} = 1$. So $N = (\mathbf{p}; \mathbf{c})$. By Theorem 27.7, $dy_1 \wedge \cdots \wedge \widehat{dy_i} \wedge \cdots \wedge dy_n((\mathbf{p}; \mathbf{a}_1), \ldots, (\mathbf{p}; \mathbf{a}_{n-1})) = \det A(1, \ldots, \hat{\imath}, \ldots, n)$, hence $\lambda(\mathbf{p}) = \sum_i (-1)^{i-1} g_i(\mathbf{p}) \det A(1, \ldots, \hat{\imath}, \ldots, n) = \sum_i g_i(\mathbf{p}) c_i = \langle G, N \rangle$.

Second proof. By linearity assume $C = M \cap \operatorname{Support} \omega$ lies in a single patch $\alpha : U \to V$ of the orientation. By Theorem 32.2,

$$\int_M \omega = \int_{\operatorname{Int} U} \alpha^* \omega = \int_{\operatorname{Int} U} \Big[\sum_{i=1}^{n} (-1)^{i-1} (g_i \circ \alpha) \det D\alpha(1, \ldots, \hat{\imath}, \ldots, n) \Big].$$

With $c_i = (-1)^{i-1} \det D\alpha(1, \ldots, \hat{\imath}, \ldots, n)$, $N(\alpha(\mathbf{x})) = (\alpha(\mathbf{x}); \mathbf{c}(\mathbf{x})/\|\mathbf{c}(\mathbf{x})\|)$ as in Corollary 38.4, and

$$\int_M \langle G, N \rangle\, dV = \int_{\operatorname{Int} U} \langle G \circ \alpha, N \circ \alpha \rangle V(D\alpha) = \int_{\operatorname{Int} U} \langle g \circ \alpha, \mathbf{c} \rangle = \int_{\operatorname{Int} U} \Big[\sum_i (g_i \circ \alpha)(-1)^{i-1} \det D\alpha(1, \ldots, \hat{\imath}, \ldots, n) \Big],$$

since $\|\mathbf{c}\| = V(D\alpha)$. ■

Lemma 38.6

Let M be a compact n-manifold in $\mathbb{R}^n$, oriented naturally. Let $\omega = h\, dx_1 \wedge \cdots \wedge dx_n$ be an n-form defined in an open set of $\mathbb{R}^n$ containing M. Then h is the corresponding scalar field,

and

$$\int_M \omega = \int_M h\, dV.$$

Proof. *First proof.* $\int_M \omega = \int_M \lambda\, dV$ with λ the value of ω on an orthonormal basis of $\mathcal{T}_{\mathbf{p}}(M) = \mathcal{T}_{\mathbf{p}}(\mathbb{R}^n)$ in its natural orientation. A patch α of the natural orientation has $\det D\alpha > 0$, so the natural orientation of $\mathcal{T}_{\mathbf{p}}(M)$ consists of the right-handed frames; the usual basis is one, and the value of ω on it is h. *Second proof.* Assume $M \cap \operatorname{Support}\omega$ lies in one patch α of the orientation. By definition $\int_M \omega = \int_{\operatorname{Int} U} \alpha^*\omega = \int_{\operatorname{Int} U} (h \circ \alpha) \det D\alpha$ and $\int_M h\, dV = \int_{\operatorname{Int} U} (h \circ \alpha) V(D\alpha)$; and $V(D\alpha) = |\det D\alpha| = \det D\alpha$ since α belongs to the natural orientation. ■

$\int_M h\, dV$ equals the ordinary integral of h over the bounded set M: $A = M - \partial M$ is open in $\mathbb{R}^n$ and the identity $i : A \to A$ is a patch of the natural orientation covering M except a set of measure zero, so by Theorem 25.4, $\int_M h\, dV = \int_A (h \circ i) V(Di) = \int_A h$, which equals $\int_M h$ since ∂M has measure zero in $\mathbb{R}^n$.

For a naturally oriented n-manifold M in $\mathbb{R}^n$, the induced orientation of ∂M behaves as in Example 34.4 for $n = 3$. The **inward normal** to ∂M at $\mathbf{p}$ is the velocity vector of a curve that begins at $\mathbf{p}$ and moves into M as the parameter increases; the **outward normal** is its negative.

Lemma 38.7

Let M be an n-manifold in $\mathbb{R}^n$. If M is oriented naturally, then the induced orientation of ∂M corresponds to the unit normal field N to ∂M that points outwards from M at each point of ∂M.

Proof. Let $\alpha : U \to V$ be a patch about $\mathbf{p} \in \partial M$ in the orientation of M, so $\det D\alpha > 0$, and let $b(x_1, \ldots, x_{n-1}) = (x_1, \ldots, x_{n-1}, 0)$. Then $\alpha_0 = \alpha \circ b$ is a patch on ∂M about $\mathbf{p}$; it belongs to the induced orientation of ∂M if n is even and to its opposite if n is odd. Let $N(\mathbf{p}) = (\mathbf{p}; \mathbf{n})$ be the normal for the induced orientation. Then $\det[(-1)^n \mathbf{n} \;\; D\alpha_0] > 0$, which implies

$$\det[D\alpha_0 \;\; \mathbf{n}] = \det\left[\frac{\partial \alpha}{\partial x_1} \;\; \cdots \;\; \frac{\partial \alpha}{\partial x_{n-1}} \;\; \mathbf{n}\right] < 0,$$

(moving the first column to the last position changes the sign by $(-1)^{n-1}$). On the other hand $\det D\alpha = \det[\frac{\partial \alpha}{\partial x_1} \cdots \frac{\partial \alpha}{\partial x_{n-1}} \;\; \frac{\partial \alpha}{\partial x_n}] > 0$. So $\mathbf{n}$ and $\partial\alpha/\partial x_n$ lie on opposite sides of the tangent plane $\mathcal{T}_{\mathbf{p}}(\partial M)$. $\partial\alpha/\partial x_n$ is the velocity vector of a curve beginning at a point of ∂M and moving into M as the parameter increases; thus $\mathbf{n}$ is the outward normal. ■

Theorem 38.8 (The divergence theorem)

Let M be a compact n-manifold in $\mathbb{R}^n$. Let N be the unit normal vector field to ∂M that points outwards from M. If G is a vector field defined in an open set of $\mathbb{R}^n$ containing M, then

$$\int_M (\operatorname{div} G)\, dV = \int_{\partial M} \langle G, N \rangle\, dV.$$

The left side is an integral with respect to n-volume, the right with respect to $(n-1)$-volume.

Proof. Let $\omega = \beta_{n-1} G$ be the corresponding $(n-1)$-form. Orient M naturally and give ∂M the induced orientation; by Lemma 38.7 the normal field corresponding to this orientation is N, so $\int_{\partial M} \omega = \int_{\partial M} \langle G, N \rangle\, dV$ by Lemma 38.5. By Theorem 31.1, $d\omega = (\operatorname{div} G)\, dx_1 \wedge \cdots \wedge dx_n$, and Lemma 38.6 gives $\int_M d\omega = \int_M (\operatorname{div} G)\, dV$. Apply Stokes' theorem. ■

In $\mathbb{R}^3$ the divergence theorem is sometimes called *Gauss' theorem*.

38.3 Stokes' theorem for 2-manifolds in $\mathbb{R}^3$

Theorem 38.9 (Stokes' theorem—classical version)

Let M be a compact orientable 2-manifold in $\mathbb{R}^3$. Let N be a unit normal field to M. Let F be a C^∞ vector field defined in an open set about M. If ∂M is empty, then $\int_M \langle \operatorname{curl} F, N\rangle \, dV = 0$. If ∂M is non-empty, let T be the unit tangent vector field to ∂M chosen so that the vector $W(\mathbf{p}) = N(\mathbf{p}) \times T(\mathbf{p})$ points into M from ∂M. Then

$$\int_M \langle \operatorname{curl} F, N\rangle \, dV = \int_{\partial M} \langle F, T\rangle \, ds.$$

Proof. Let $\omega = \alpha_1 F$ be the corresponding 1-form; by Theorem 31.2, $\operatorname{curl} F$ corresponds to the 2-form $d\omega$. Orient M so that N is its unit normal field; by Lemma 38.5, $\int_M d\omega = \int_M \langle \operatorname{curl} F, N\rangle \, dV$. If ∂M is non-empty, its induced orientation corresponds to the unit tangent field T (Example 34.5: (N, T, W) is right-handed, i.e. $W = N \times T$ points into M), so Lemma 38.1 gives $\int_{\partial M} \omega = \int_{\partial M} \langle F, T\rangle \, ds$. Apply Stokes' theorem. ■